

Membership inference recovers most patients from synthetic continuous glucose monitoring, and removing one representational capacity returns it to chance

Ling Cheng^{1*}

^{1*}TODO, Imperial College London, London, United Kingdom.

Corresponding author(s). E-mail(s): lcheng2@ic.ac.uk;

Abstract

Synthetic continuous glucose monitoring data is increasingly deposited in public repositories to enable sharing, on the premise that a release containing no copied record cannot identify anyone. We tested that premise directly. Using a public multi-study cohort of 506 people we trained, for each of 26 target patients, one generator on a background population and one on that population plus the target, and asked whether the patient’s real recordings lay closer to the release that had seen them. The symmetric design supplies its own null: because the background excludes every target, each of the other 25 one-patient-different generators is itself a legitimate non-member model for any given target, giving 25 draws of the statistic under the null and an empirical significance per patient without further training. Across seven generators and three data compositions, a construction that memorises by definition is detected for 26 of 26 patients, confirming full power; the strongest published generator exposes 14 to 20 of 26. A generator whose latent bottleneck cannot store the absolute level of a window per timepoint exposes 1 to 3, against the 1.3 expected by chance, while achieving the best generation fidelity of any learned model tested. Risk is not confined to patients with atypical glycaemic profiles: it reaches the matched controls equally.

Keywords: synthetic data, membership inference, continuous glucose monitoring, privacy, generative models, type 1 diabetes

1 Introduction

Continuous glucose monitoring produces one of the densest records in routine care: a value every five minutes, indefinitely, for millions of people living with diabetes. It is also among the hardest records to share. A trace carries sleep, meals, work patterns and illness alongside glycaemia, and the consent under which it was collected rarely anticipates secondary use. Synthetic data is the most widely proposed way out of that bind — release a generated cohort that reproduces the statistics of the real one and share it without restriction — and the proposal is now moving from argument to obligation, with journals and funders increasingly requiring that a synthetic dataset be deposited publicly, with a persistent identifier and an open licence, as a condition of publication. TODO — citations for the clinical value of CGM, for synthetic data as a sharing route, and for the deposition requirements.

The privacy of such a deposit is usually argued rather than measured. The common arguments are that no real record appears in the release, or that the distance from each synthetic record to its nearest real neighbour exceeds some threshold. Neither is a statement about what an adversary can infer. Membership inference — asking whether a particular person’s data was in the training set — is the standard formulation of that question, and it has been studied extensively for classifiers and for images, but far less for generative models of physiological time series, where the attacker sees only released samples and never the model. Continuous glucose monitoring has attracted particularly little attention, partly because glucose is homeostatically bounded and a trace is widely assumed to be too constrained to identify anyone. TODO — citations for membership inference, for its application to generative models, and for existing privacy evaluations of synthetic health data.

Where privacy of synthetic health data has been evaluated, it has usually been summarised as a single number for a whole cohort. That is precisely the summary that hides the patients who matter. Rare physiology is simultaneously the thing a shared dataset exists to preserve — brittle control, frequent hypoglycaemia, an unusual response to insulin — and the thing most likely to make its owner identifiable. A generator that is safe on average may still expose the people whose data motivated the release, and an aggregate metric will not say so. This is the sparse-population fairness question, and answering it requires attacking individual patients rather than scoring a distribution. We measure it directly. Using a public multi-study cohort of 506 people, we train a generator on a background population and 26 further generators that each add exactly one target patient, so that a pair of models differs by one person and nothing else, and we ask whether that person’s real recordings lie closer to the release that saw them. The design supplies its own null: because the background excludes every target, each of the other 25 one-patient-different generators is a genuine non-member model for any given target, giving 25 draws of the statistic under non-membership and an empirical significance per patient at no additional training cost. Across seven generators and three data compositions we find that a construction which memorises by definition is detected for every patient, that the strongest published generator is detected for most of them, that removing one representational capacity from the generator returns this to the chance rate without any cost in fidelity, and

that the exposure is not confined to the patients with atypical physiology the study was built around.

2 Results

2.1 The design supplies its own null, and the attack detects a memoriser in every patient

For each target patient we train one generator on a background population and one on that population plus the target, and measure how much closer the patient’s real windows lie to the release that saw them (Fig. ??). Reading that gap requires knowing what it does when membership is absent, and the design answers this without further training.

The background set excludes *all* 26 targets. For a given target, each of the other 25 one-patient-different generators is therefore a legitimate *non-member* model: it saw its own target, not this one. Evaluating the same statistic against those 25 gives its distribution under the null, and the observed value’s rank among them is an empirical significance for that patient. This also removes a bias that a single shared reference introduces. All 26 comparisons in the published convention share one background release, so that release’s own idiosyncratic draw enters every comparison with the same sign; replacing it with the mean over the 25 non-member models deletes that offset structurally. On the seven-day composition the offset is large enough to change the conclusion on its own: the shared-reference gap is -0.0018 , and the same data read against the non-member mean is $+0.0003$.

Against this null a construction that memorises by definition — resampling its own training windows — is detected for **26 of 26 patients** on every composition. The attack has full power where memorisation is total.

We do not use a within-window shuffle as a floor. Permuting the samples inside a window destroys temporal order but preserves that window’s value distribution exactly, and identity in glucose traces resides largely in that distribution: under such a permutation the most identifiable patient in this cohort still scores 1.000. A control that leaves the signal intact is not a control.

2.2 Membership is recoverable for most patients from the strongest published generator, and not only for atypical ones

Table 1 reports, for every generator and composition, how many of the 26 patients show membership significant at the 5% level against the null above. With 26 patients, 1.3 is the number expected by chance.

The strongest published generator in our comparison exposes **18 of 26** patients on one-day glucose, **14 of 26** with basal insulin added, and **20 of 26** on seven-day glucose. This is an order of magnitude above chance and is not a threshold artefact: the same release also carries the largest median gap in the table.

The risk is not concentrated in patients with atypical glycaemic profiles. The design pairs 13 such patients against 13 controls matched on recording duration, and the

Table 1 Generation fidelity and membership exposure for seven generators on three data compositions. Context-FID is lower-is-better. “Exposed” counts patients of 26 whose membership is significant at the 5% level against the non-member null of Section 2.1; **1.3 of 26 is the number expected by chance**. Copy-paste is a construction that resamples its own training windows and is included to show the attack’s power, not as a candidate release.

Generator	1 day, glucose		1 day, glucose + basal		7 days, glucose	
	FID	Exposed	FID	Exposed	FID	Exposed
Copy-paste (ceiling)	0.034	26	0.043	26	0.073	26
This work	0.038	1	0.055	3	0.127	3
Backbone, unmodified	—	1	0.055	1	—	—
DiM-TS	0.095	18	0.159	14	0.326	20
FourierDiffusion	0.100	1	0.194	1	—	—
DiffWave	0.368	0	0.491	0	—	—
Diffusion-TS	0.422	3	2.058	1	—	—

Fidelity values are being re-scored, one call per composition, so that a column is internally comparable; the same release re-scored in a different compute job differs by one to five per cent. Exposure counts are final. Dashes are cells not trained: only three generators were taken to seven days.

significant patients divide almost evenly between them — 8 atypical against 10 controls on one-day glucose, 7 against 7 with insulin, 8 against 12 at seven days. Tested directly, the between-arm contrast does not separate on any composition (AUC 0.50–0.64, all $p > 0.2$). We designed this study to detect whether rare physiology carries extra membership risk and, by this measure, it does not: what is unusual about these patients makes them no easier to recover than a typical patient from the same cohort.

TODO — state the median gaps per arm from `results/matrix/membership_null.json` once the re-scored fidelity column lands, and give Table 1 its final numbers.

2.3 The membership signal is carried by absolute glucose level and by temporal ordering

TODO — ALL COUNTS IN THIS SECTION ARE VOID and are being recomputed by `scripts/mechanism_null.py` against the non-member null. The ablation itself (destroy one coordinate, repeat the attack) is sound; what is void is reading the result against the within-window shuffle. Two specific claims must not be carried over: that “removing level and ordering together reaches the floor”, which has no floor to reach, and that differencing shows level carries the signal — a pure level removal costs only 24 to 21 patients on the old reading, so differencing’s effect came mostly from the low-frequency structure it also destroys. To locate the signal we destroyed one coordinate of the released windows at a time and repeated the attack unchanged (Fig. ??). Removing within-hour detail, by averaging to hourly resolution, costs nothing: the strongest leaking release still exposes **23 of 26** patients with a median AUC of 0.692, against 24 and 0.653 untouched. Whatever is being recovered, it is not fine structure.

Two coordinates do carry it. Differencing the windows, which removes absolute level while preserving shape, takes exposure from **24 to 13 of 26**. Sorting each window, which destroys temporal ordering while preserving the value distribution exactly, takes it to **13 of 26** as well. Neither alone reaches the floor.

Destroying both does. With level and ordering removed together, exposure falls to **7 of 26** at a median AUC of 0.487 — the permutation floor for this release is 7 at 0.503. *The two coordinates jointly account for the entire recoverable signal*; nothing measurable remains once they are gone.

The two are not interchangeable with respect to who is protected. Removing temporal ordering leaves the atypical arm separated from the control arm (median AUC 0.600 against 0.476), whereas removing absolute level collapses that separation (0.568 against 0.531). Level is thus the coordinate that carries the specifically atypical part of the signal, which is what motivates the intervention in Section 2.6.

On the two-channel composition, level alone is sufficient: differencing takes exposure from 15 to **6 of 26** against a floor of 8, at or below chance without any need to touch the ordering.

2.4 Adding basal insulin lowers measured exposure while lowering fidelity, and the two cannot be read apart

TODO — REWRITE AGAINST THE NEW NULL. The previous version of this section claimed that adding basal insulin protects typical patients while leaving atypical ones exposed (13 of 13 controls falling to 5 of 13, against 11 of 13 atypical falling to 10). Those counts came from a threshold read against the within-window shuffle, which Section 2.1 retires. Against the non-member null the split is 7 atypical and 7 controls, i.e. no asymmetry, and the claim does not survive.

What does survive, and should be the content of this section: exposure falls from 18 to 14 of 26 when basal insulin is added, while *every* generator fits the two-channel composition worse. So the apparent protection is bought by degrading the release, not by any privacy mechanism. The tension to state is that basal is the more identifying channel in the real recordings — one day of basal alone retrieves its subject 29.3% of the time against 1.5–2.6% for glucose — so an intrinsically more identifying channel lowered measured exposure. Figure ?? must carry real-data identifiability, exposure, and fidelity together; any two of them support the wrong conclusion.

2.5 Removing the model capacity that stores absolute level returns membership recovery to the chance rate

TODO — re-express against the non-member null. The fidelity and the significance counts (1, 3 and 3 of 26 against 1.3 expected by chance) are in Table 1 and stand. What must go is every phrase of the form “at the permutation floor”, and the motivation paragraph, which rests on Section 2.3 still naming absolute level as the carrier — that is being re-measured. Section 2.3 names absolute level as the carrier. We therefore changed the generator so that level cannot be stored per timepoint, rather than adding a penalty that discourages storing it (Fig. ??).

Two changes were made. First, the training task masks whole coordinates instead of scattered cells: contiguous blocks of time, or the level of the whole window. Scattered-cell masking is solvable by interpolating between neighbours, and interpolation is exactly the local correlation in which identity resides. Second, the channels of the latent bottleneck that carry level are pooled along time and broadcast back, so that this part of the representation takes a single value for the entire window and is *structurally unable* to hold a per-timepoint trace. The capacity is removed, not discouraged.

The released sets show no recoverable membership signal. The median gap is **0.0026** on the glucose-only composition and **0.0005** with basal insulin, and in neither composition is it significantly positive in either arm ($p = 0.064$ and 0.682 ; $p = 0.271$ and 0.249) — against $p < 0.001$ for the strongest learned baseline. Per patient, exposure is **10 against a floor of 10** and **5 against a floor of 5**. Fidelity is not paid for this: at Context-FID **0.038** and **0.055** the generator is the best learned model in Table 1 on both compositions, second only to the copy-paste ceiling.

The mechanism behind the masking change can be read directly from training. The task that withholds absolute level should be harder than the original scattered-cell task if the model is genuinely being forced to generate level rather than to copy it. It is, in every setting we trained, but only by **12–24%**. The contiguous-block task is harder by **2–5×**. Most of the effect of this change therefore comes from forcing the model to generate temporal order, not from withholding level, and we report it that way.

TODO — state that the companion group-decorrelation loss cannot constrain the quantity the pooling preserves, since it z-scores along time and thereby removes exactly the window mean that the pooling broadcasts. The pooling does the work.

2.6 The released windows reproduce twenty clinical glycaemic metrics that the strongest baseline distorts

TODO — the privacy half of this section is VOID. It quoted exposure counts read against the within-window shuffle, which Section 2.1 retires. The clinical half (twenty metrics, translation-invariant ones exactly unchanged, scale-dependent ones destroyed by stripping and restored by the permutation re-offset) stands as measured and needs no change. Fidelity measured in an embedding space does not establish that a release is clinically usable. We therefore computed twenty established glycaemic metrics per window and compared their distributions against the real cohort (Table ??).

The released windows reproduce the real distributions across every family of metric. Mean amplitude of glycaemic excursions is **51.22** against a real **51.23**; glucose variability percentage **42.44** against **42.79**; CONGA-1 **41.83** against **41.69**; time in range **74.3%** against **74.0%**; and time below 54 mg/dL **0.55%** against **0.58%**.

The strongest learned baseline does not. It understates short-term variability by a wide margin — mean amplitude **38.28**, a **25%** shortfall, and glucose variability percentage **26.81**, a **37%** shortfall — and it understates exposure to clinically significant hypoglycaemia by **45%** (**0.32%** against **0.58%** of time below 54 mg/dL). A cohort built from that release would look less variable and safer than the patients it was derived from, which is the direction that matters for a dataset intended to support algorithm development.

A release-time intervention for custodians who cannot retrain.

Section 2.3 identified absolute level as the coordinate that carries the specifically atypical part of the signal. A custodian can remove it from an already-trained generator’s output: publish each window with its own level subtracted, and restore a level drawn by permutation from the levels of the other released windows. The permutation preserves the marginal distribution of levels across the release exactly, while severing which level belongs to which curve.

We measured what this costs clinically, for each of the twenty metrics, as the Kolmogorov–Smirnov distance between the released distribution and the real one (Table ??). The metrics fall into two groups, and the split is by construction rather than by luck.

Metrics defined on shape are *exactly* unchanged, because subtracting a constant from a window does not move them: mean amplitude of glycaemic excursions stays at KS 0.040, CONGA-1 at 0.017, standard deviation at 0.016, MAGE at 0.026, glucose variability percentage at 0.046, and the lability index and average absolute rate of change likewise do not move by a single digit.

Metrics defined on the value scale are destroyed by stripping alone and *restored by the re-offset*. Time below 54 mg/dL moves from KS 0.006 in the untouched release to **1.000** once the level is removed — a complete loss — and back to **0.192** once a level is restored. Time in range goes 0.012, 0.977, **0.126**. Mean glucose goes 0.022, 1.000, and **0.022**: exactly back, because a permutation leaves the marginal untouched. Level-stripping alone is therefore not a usable release; level-stripping with re-offsetting is.

What the re-offset does not restore, and cannot, is the association between a patient’s glycaemic shape and the level they sit at. That association is the privacy gain and it is also a genuine loss of utility: a study asking whether patients with greater variability run higher mean glucose cannot be run on such a release, and the release must say so.

TODO — add the privacy side of this exact transformation, now computing, as a column of Table ?. The numbers in Section 2.3 are for differencing, which is a different transformation, and must not be reused here.

TODO — Table ??: twenty metrics $\times$ (released, level-stripped, re-offset) $\times$ (median, interquartile range, KS, Wasserstein), built from `results/clinical/after_strip.json`.

3 Discussion

Synthetic continuous glucose monitoring data is released on the premise that a record which was never copied cannot identify anyone. The measurements here do not support that premise. Against a null the design supplies for free — the 25 other one-patient-different generators, each a genuine non-member for any given target — membership was recoverable at the 5% level for 18 of 26 patients from a one-day glucose release by the strongest generator we compared, 14 with basal insulin added, and 20 at seven days. The chance expectation is 1.3. The same instrument detects a deliberate memoriser in 26 of 26 patients, so the power is not in question.

The risk is not the one we designed the study to find. The cohort was built around 13 patients with atypical glycaemic profiles, matched to 13 controls, on the expectation that rare physiology would be the vulnerable group. It is not. Significant patients divide almost evenly between the arms in every composition, and the between-arm contrast does not separate anywhere (AUC 0.50–0.64, all $p > 0.2$). A custodian releasing synthetic CGM should not expect the exposure to be confined to unusual patients, and a privacy audit that samples only the unusual ones will under-report it.

TODO — the mechanism paragraph. Rewrite once `scripts/mechanism_null.py` reports which coordinate carries the signal against the non-member null. Do not reuse the previous reading: a pure removal of absolute level was far weaker than differencing, so the earlier attribution to level conflated it with the low-frequency structure differencing also destroys.

A generator whose latent bottleneck cannot store the absolute level of a window per timepoint — capacity removed rather than penalised — is recovered for 1, 3 and 3 of 26 patients across the three compositions, which is the chance rate, at the best generation fidelity of any learned model we tested. Three of the four published baselines are also near chance, but at four to ten times worse fidelity; being unrecoverable is not by itself difficult, and the result here is the combination.

TODO — position against prior work on privacy-preserving synthetic health data, differential privacy for generative models, and nearest-neighbour privacy heuristics; citations needed.

Three limitations bound what this supports. First, one replicate: the design trains one background generator and 26 paired models, and while the non-member null makes each patient’s significance interpretable within that replicate, the between-arm comparison remains a screen rather than a test. Second, the intervention removes what is partly the clinically useful signal — mean glucose is both a primary clinical quantity and a carrier of identity — and while a permutation re-offset restores its population distribution, it permanently severs the association between a patient’s glycaemic shape and the level they sit at, so a release built this way cannot support a study of that association and must say so. Third, “atypical” here is defined by one family of distributional metrics on one cohort; that the arms did not separate is evidence about this definition, not about every notion of an unusual patient.

A note on controls, since it changed our own conclusions twice. An earlier version of this analysis used a within-window permutation as its null. That transform destroys temporal order but preserves each window’s value distribution exactly, and identity in glucose traces lives largely in that distribution — under it, the most identifiable patient in this cohort still scores 1.000. Every exposure count we had read against it was a comparison between two quantities that were never ordered. The null a privacy evaluation uses deserves the same scrutiny as the attack.

For a custodian preparing a deposit: evaluate membership per patient against a null built from models that genuinely did not see that patient, which a leave-one-out design supplies at no extra training cost; do not read a lower aggregate leakage number as protection without checking the fidelity that produced it; and do not assume the exposure is confined to the unusual patients in the cohort.

4 Methods

4.1 Sex as biological variable

The source release records a single self-reported **gender** field and no separate biological-sex variable. Across the 506 subjects of the cohort it takes the values female ($n = 274$), male ($n = 231$) and non-binary ($n = 1$). This field was not used at any point: it does not enter the definition of atypical profiles, the matching of controls, the composition of the background set, or any model input, and no result reported here is stratified by it. Whether membership risk differs by sex is therefore an open question that this study is not powered to answer — with 13 targets per arm, a sex-stratified contrast would rest on fewer than seven subjects per cell.

4.2 Cohort and data acquisition

Recordings were drawn from a public multi-study continuous glucose monitoring release comprising 1,291 participants across twelve source studies. From these we retained the 506 subjects with at least 30 complete recording days on the glucose channel, giving 6,072 windows. The same subject set is used for both data compositions, so that the one-channel and two-channel results differ only in what is released and not in who is in the cohort.

4.3 Window construction and subject key

Recordings are sampled every five minutes. A window is one gap-free calendar day of 288 consecutive samples; partial and interrupted days are discarded rather than imputed, so no window contains interpolated values. Values are standardised per channel and clipped at five standard deviations; the glucose channel’s constants are a mean of 145.94 mg/dL and a standard deviation of 57.65 mg/dL, and all clinical quantities in this paper are computed after inverting that transformation.

Subject key.

Participant identifiers in the source release are unique only *within* a source study: 241 of 1,291 identifiers (18.7%) recur across two or more studies, and one identifier spans five. The subject key used throughout is therefore the pair (source study, participant identifier). Keying on the identifier alone merges distinct people into composite subjects, which in a membership design means holding several people in or out as a single unit.

4.4 Definition of atypical glycaemic profiles

Atypical subjects were identified by consensus across thirteen distributional and variability metrics. For each metric, subjects in the top five per cent were flagged; a subject flagged by at least seven of the thirteen metrics was provisionally called atypical. The whole procedure was repeated under four random seeds and only subjects selected under all four were retained, giving 13 atypical subjects. *The threshold of seven votes*

is fixed for comparability with earlier work on this cohort and is not read off a mode in the vote histogram, which is unimodal here.

4.5 Control matching

Each atypical subject was paired with one control drawn from the 493 non-atypical subjects and matched on recording duration, since the number of released windows for a target scales with it and nearest-neighbour distance falls with the size of the released set. Matching was exact or near-exact: median relative gap in recording days was below one per cent and no pair exceeded tolerance. Controls were not matched on any glycaemic quantity, so the control arm remains a sample of ordinary physiology rather than a shape-matched twin of the atypical arm.

4.6 Paired membership-inference design

For each of the 26 targets we trained one generator on a background set of 475 subjects (5,741 windows) that contains no target, and one on that background plus the single target. Both members of a pair share the generator, its hyperparameters and its training seed; they differ by one subject and by nothing else. The background generator is shared across all 26 pairs, so the design is symmetric: both arms are membership pairs, and neither is a non-membership control. Each model releases a number of synthetic windows equal to its own training-set size, which is what a custodian releasing a dataset would do.

4.7 Generative models and training budgets

Seven arms were evaluated: a copy-paste construction that resamples its training windows with replacement, four published generators for physiological time series, the unmodified backbone of our own model, and the backbone with the two structural changes of Section 2.5. TODO — name each published generator, its citation, its capacity and its training budget, and state that all arms were trained to the same budget on this cohort.

4.8 Membership statistic

For a target t with real windows R_t , let d_{IN} be the distance from R_t to the release of the generator that saw t , and d_{OUT} the distance to the release of the generator that did not. The statistic is the gap $d_{\text{OUT}} - d_{\text{IN}}$, positive when the target’s own windows lie closer to the release that was trained on them. Distances are Euclidean over the flattened window, reduced across the released set by its minimum and across the target’s windows by their mean. *This variant was fixed on the copy-paste arm and frozen before any learned generator was attacked*, so that the test is not selected on the outcome.

Matching the released-set size.

A generator trained with one additional subject also releases that subject’s windows, roughly 0.1% more than its paired background model. Nearest-neighbour distance

falls with the size of the released set, so this alone would bias the gap upwards. Both releases are therefore cut to the same size, at random and with a fixed seed, before any distance is computed.

4.9 Null distribution for the membership statistic

The background generator is trained on subjects that exclude *every* target in the design. For a given target, each of the other 25 one-patient-different generators therefore saw a different patient and not this one, and is a legitimate non-member model for it. Evaluating the same statistic against those 25 releases gives its distribution under the null of non-membership, and the observed value’s rank among them is a one-sided empirical p -value for that patient, with a resolution of $1/26$. No additional training is required.

The same construction removes a bias. Under the paired convention all 26 comparisons share one background release, so that release’s own draw enters every comparison with the same sign and the median gap partly estimates it rather than membership. Replacing the single reference with the mean over the 25 non-member models deletes the offset structurally. On the seven-day composition it is large enough to invert the sign of the result.

We do not use a within-window permutation as a null. Permuting samples within a window destroys temporal order but preserves that window’s value multiset exactly, and identity in glucose traces resides largely in the value distribution rather than in its ordering; the most identifiable patient in this cohort scores 1.000 under that transform. Earlier revisions of this work used it, and every count read against it was uninterpretable.

4.10 Generation fidelity metrics

Fidelity is reported as Context-FID: a TS2Vec encoder is trained on the real half, real and released windows are embedded to full-series representations, and the Fréchet distance between the two representation sets is taken. We also report a discriminative score, the balanced accuracy of a recurrent classifier trained to separate real from released windows, for which 0.5 is indistinguishable. *Scores for a given release are not exactly reproducible across compute jobs*; on byte-identical inputs with identical seeds we observe differences of one to five per cent, so quantities that are compared with one another were computed in a single job.

4.11 Glycaemic utility metrics

Twenty established metrics of glycaemic variability, hypoglycaemia, hyperglycaemia and overall control were computed per window on the inverted mg/dL scale. Because these metrics are defined per recording rather than per cohort, we compare *distributions* rather than means: each metric is summarised by its median and interquartile range across windows, and the distance between a released distribution and the real one is reported as both a Kolmogorov–Smirnov statistic and a Wasserstein distance. TODO — cite the source of the metric definitions and their discriminative ratios, and note the deviation from its twelve-day windows.

4.12 Statistical analysis

Per-patient membership scores are Mann–Whitney statistics of the two distance distributions, normalised to the unit interval. The between-arm contrast is a Mann–Whitney test of the atypical arm’s gaps against the control arm’s. *All 26 pairs share one background generator, so their gaps are correlated*; within a single replicate the resulting p -values are a screen rather than a test, and the corresponding test is the spread of the arm difference across independent replicates. **This study reports one replicate**, and conclusions that depend on the between-arm contrast alone are qualified accordingly in the Discussion.

Data availability. TODO — must name the deposited synthetic dataset, its repository, its persistent identifier and its licence. The collection requires this at submission.

Code availability. The code used to generate the results presented in this study is publicly available at https://github.com/Cranooooooooo/Metabonet_CGM_MIA.

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Supplementary information. TODO.

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